```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow

# Upload image
uploaded = files.upload()

# Get uploaded file name
filename = next(iter(uploaded))

# Read image in grayscale
img = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)

# Check if image loaded correctly
if img is None:
    print("Error: Image could not be loaded.")
else:
    # Sobel Y edge detection
    sobel_y = cv2.Sobel(
        img,
        cv2.CV_8U,
        0, 1,
        ksize=5
    )

    # Display original image
    print("Original Image:")
    cv2_imshow(img)

    # Display Sobel Y result
    print("Sobel Y Edge Detection:")
    cv2_imshow(sobel_y)

    # Save output
    cv2.imwrite("sobel_y.jpg", sobel_y)

    print("Sobel Y image saved as sobel_y.jpg")
```

Choose Files sampled_image.jpg

sampled_image.jpg(image/jpeg) - 9699 bytes, last modified: 7/22/2026 - 100% done

```
Saving sampled_image.jpg to sampled_image (3).jpg
Original Image:
```

```
Sobel Y Edge Detection:
```

```
Sobel Y image saved as sobel_y.jpg
```